ASSIGNMENT-1

TOPIC: ANALYSING EXTREME EVENTS AND DISASTERS IN INDIAN STATES

DISASTER COVERED: JAMMU AND KASHMIR FLOODS 2014

COURSE NAME:

EXTREME EVENTS AND DISASTERS

COURSE CODE: ECCSS 11

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1. INTRODUCTION

Floods are among the most frequent and catastrophic natural disasters occurring around the globe (Berz et al. Citation2001) and seriously impacting human lives and infrastructure. Due to poor socio-economic conditions, inadequate warning systems and preparedness, floods have a greater impact in the developing countries. Natural hazards, in the form of floods and debris flows, are a major threat to the people living in the high mountain regions (Watson & Haeberli Citation2004). This led me to choose the ***2014 Jammu and Kashmir Floods*** as the disaster for my analysis in this report.

In September 2014, the Jammu and Kashmir region of India experienced one of the most devastating flood events in its history, often referred to as "***The Great Flood***." This catastrophic event was primarily triggered by continuous heavy rainfall from September 3 to September 7, which coincided with the last stages of the southwest monsoon. The floods resulted in widespread devastation across the Kashmir Valley, affecting both urban and rural areas.

The incessant rains led to the Jhelum River overflowing its banks, inundating approximately 853 km² of land, including residential areas and agricultural fields (Romshoo et al., 2018). The flooding was particularly severe in Srinagar, where around 100 km² of the city was submerged, affecting 998 mohallas and colonies across 59 wards (Malik & Hashmi, 2020). The floods displaced over one million people, with more than 3,000 villages affected (Carpenter et al., 2020). The human toll included numerous casualties and significant disruptions to daily life.

Jammu and Kashmir is a mid-size state situated in the northern part of India. According to the 2011 state census, there are 6,651 villages, 104 urban areas and 8 urban agglomerations in the state. These villages and towns are distributed in 22 districts and 82 sub-districts. The population of the state was recorded at 1.25 crores or 12.25 million. The floods directly affected more than 2,600 villages in the state and submerged 30 per cent of the urban areas. Out of the 2,600 villages, almost 400 villages were completely submerged and 2,225 partially submerged with more than 300 villages completely cut off (inaccessible). In the urban areas, the water levels rose up to 20 feet whereas many villages were cut off due to the destruction of the only bridges and roads leading into these villages. The continuous bad weather made rescue operations almost impossible for the first few days and the government struggled to make an appearance to help the people.

In terms of districts, 10 out of the 22 districts were badly hit with the districts in South Kashmir experiencing severe devastation. Anantnag, Kulgam, Shopian, and Pulwama (all South Kashmir) districts were completely inundated and rendered inaccessible. More than 5 million people were severely affected by the catastrophic event while electricity and communication were completely shut down making it nearly impossible to coordinate, evacuate and rescue.

According to the Centre for Science and Environment (CSE), India, the floods in Kashmir follow a recognizable pattern of heavy rainfalls that was previously seen in other parts of India: Mumbai (2005), Leh (2010) and Uttarakhand (2013). The meteorological (MET) department confirmed that they had issued a warning regarding heavy rainfall and urged people to move to higher ground, a warning that most people did not heed, and the state government did not do anything about. This warning should have been given more weight considering the recent floods in Uttarakhand. To be more precise as to what caused the floods, CSE indicated that the floods were caused by a combination of intense rain, mismanagement, unplanned urbanization and a lack of preparedness. Local scientists, Humayum Rashid and Gowhar Naseem, had indicated in 2008 that there was a breakdown of the natural discharge system in the valley that could prevent water from flowing out of the valley. In their paper from the 12th World Lake Conference, the scientists argued that the natural discharge system had collapsed due to the degradation of the network of lakes.

The economic consequences were staggering, with estimated losses exceeding INR 1 trillion (approximately USD 15 billion) (Yaseen, 2014; ASSOCHAM India, 2014). Key sectors such as agriculture, trade, infrastructure, and tourism suffered immensely. For instance, the agricultural sector alone faced losses amounting to INR 3,674 crore, with substantial damage to crops including a reported loss of INR 1,000 crore to the apple industry (ASSOCHAM India, 2014). The flood highlighted significant vulnerabilities within the region's socio-economic fabric. Areas such as Srinagar were particularly hard-hit due to their geographical location and urban density.

1.1.Summary of the Event

In early September 2014, unprecedented rainfall led to severe flooding in the Kashmir Valley, particularly affecting the Jhelum River basin. The floods were among the worst in six decades, causing extensive damage to infrastructure, homes, and agriculture.

1.2. Districts Affected

Srinagar, Bandipora, Pulwama, Baramulla, Budgam, Anantnag, and Shopian.

1.3. Timeline

Flooding began around ***September 3, 2014***, with peak inundation occurring between ***September 5*** and ***8***.

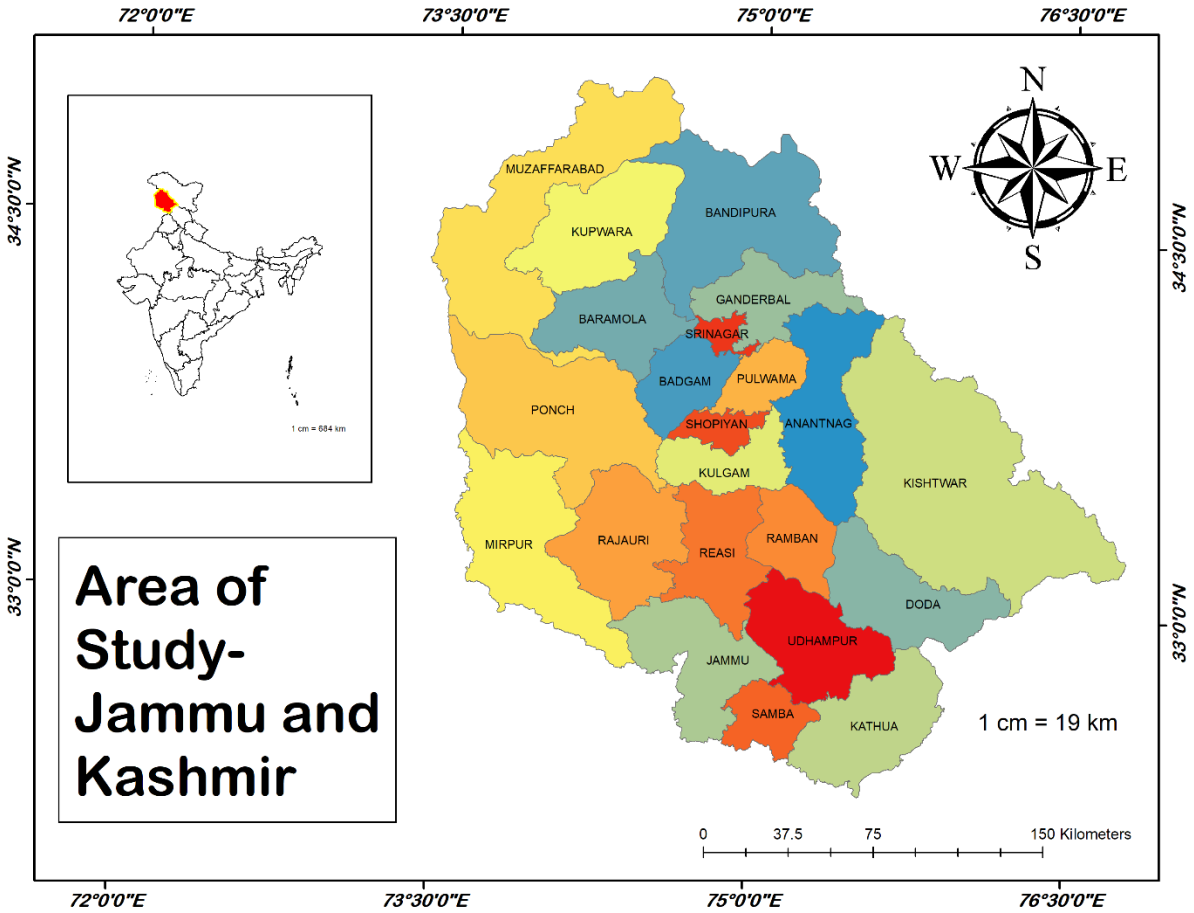
1.4. Facts and Figures

- Approximately ***557 km²*** of the Kashmir Valley was inundated.
- The ***Jhelum River*** exceeded its danger mark, leading to breaches in embankments.
- Approximately ***1.9 million*** people were affected.
- At least ***474 fatalities*** reported.
- Significant damage to homes, roads, and public infrastructure

2. METHODOLOGY

2.1. Area of Study

My study is focussed across all the districts of the present-day Jammu and Kashmir post separation of Ladakh as a Union Territory.



Map 1. Study area map

2.2. Data Collection and Preprocessing

The data for all the analysis in this report has been restricted to:

- The IMD's Gridded Annual Rainfall Data 2014
- Landsat 8 data retrieved from USGS Earth Explorer
- The DEM file for India
- India State Boundary shapefile
- India Districts shapefile

To begin with the data for Jammu and Kashmir was clipped as a separate shapefile from India Districts shapefile. Similarly, the DEM for India was clipped to Jammu and Kashmir as its extent.

The Landsat data was the most difficult part of this assignment. Due to the flood being induced because of heavy rainfall during the period, all data available had a lot of cloud cover. This was creating errors in the data analysis. To analyse the NDWI, the districts that had very low cloud cover were clipped and the NDWI was calculated for these districts only.

The data processing included:

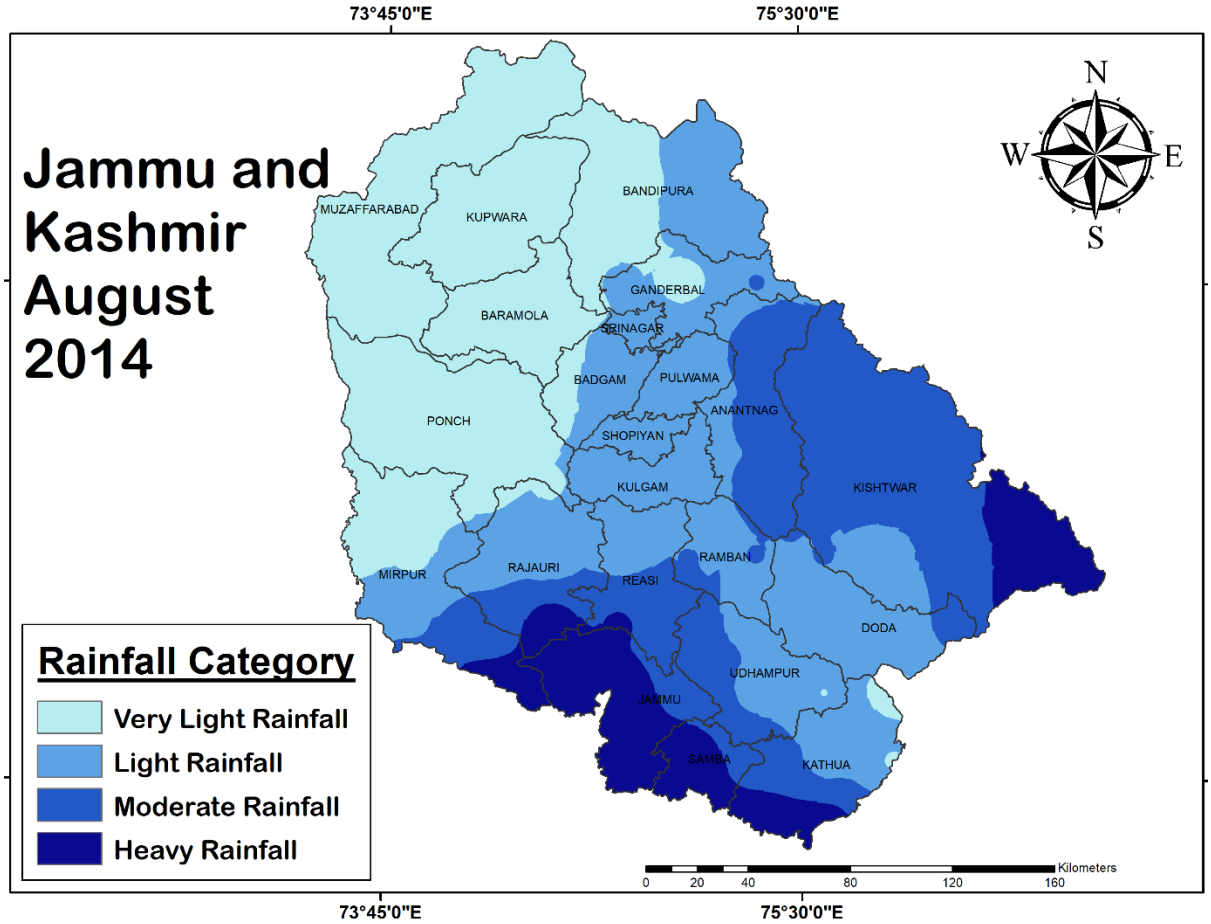
- Interpolation of Rainfall data for August, September and October.
- Pre-flood and Post-flood NDWI estimation.
- Topographical analysis in the form of Aspect, Slope, Elevation, Contour and Hill shade.

The Spatial Analysis using GIS Tools includes:

- Fishnet creation
- Zonal statistics

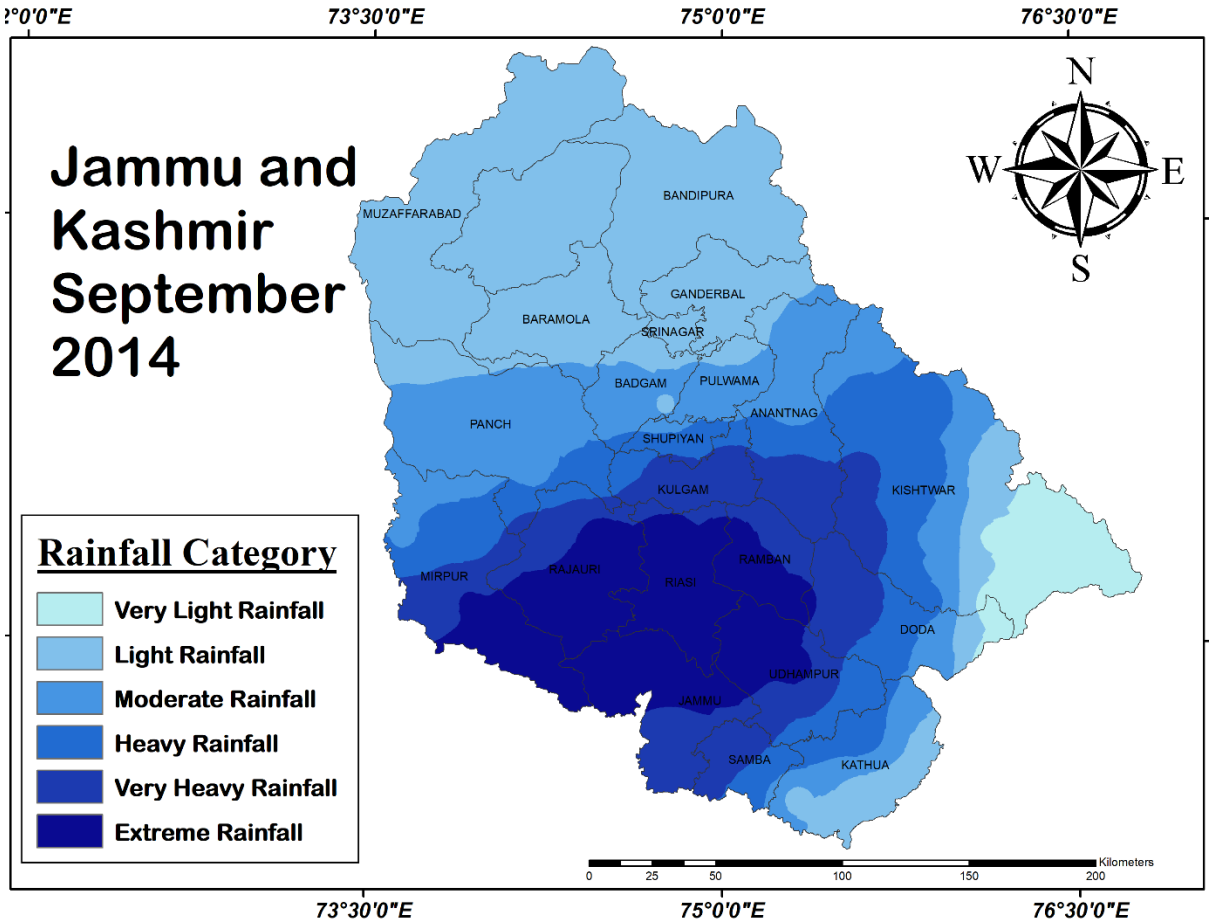
3. Spatial analysis and Interpretation

3.1. Precipitation analysis



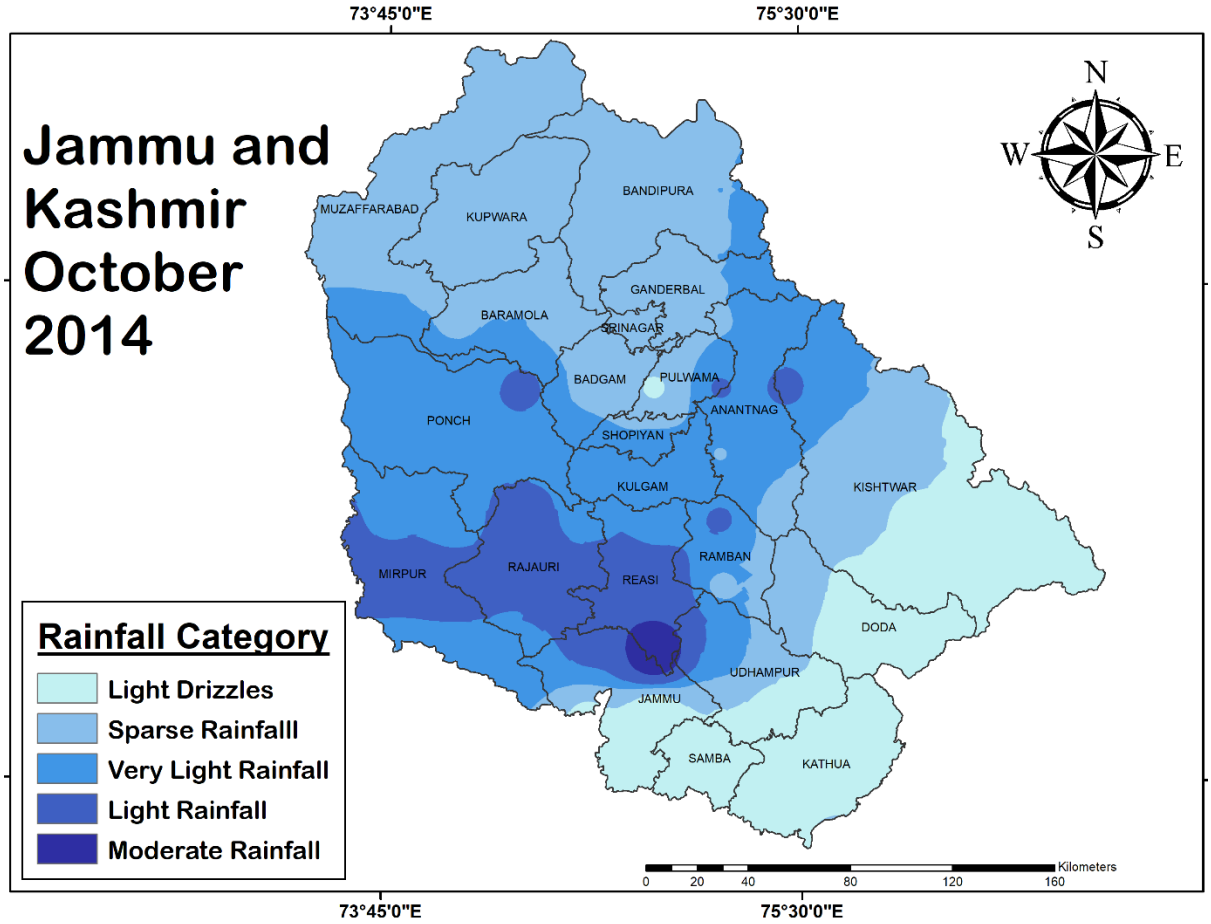
Map 2. Average rainfall in August (2014)

In August 2014, the rainfall distribution across Jammu and Kashmir exhibited a pronounced west–south bias. Western Jammu districts—Rajauri, Mirpur, Jammu, and Samba—experienced **heavy** downpours, while adjacent zones such as Kulgam, Anantnag, Kishtwar, Doda, and Udhampur reported **moderate** rainfall. In contrast, central Kashmir districts (Srinagar, Budgam, Pulwama) fell into the **light** category, and the far northern highlands (Kupwara, Baramulla, Bandipora, Muzaffarabad, Poonch) saw only **very light** rains. This early moisture loading in the Pir Panjal foothills likely primed soils for the more extreme events to come.



Map 3. Average rainfall in September (2014)

September's monsoon peak brought a dramatic spike in rainfall intensity, concentrated in the south–central catchments of the Jhelum basin. Reasi and Jammu recorded **extreme** rainfall, while Rajauri, Mirpur, and Samba registered **very heavy** values. Ramban and Kulgam similarly fell into the **very heavy** class, with Kishtwar, Anantnag, and Udhampur seeing **heavy** rains. Meanwhile, northern districts remained relatively dry—Kupwara, Baramulla, and Bandipora again saw only **very light** showers, highlighting the south-central focus of the deluge that ultimately breached embankments and inundated the valley.



Map 4. Average rainfall in October (2014)

By October, monsoon withdrawal was well underway. High-altitude Kashmir (Kupwara, Bandipora, Baramulla, Ganderbal, Srinagar, Budgam, Pulwama) experienced **light drizzles** or **sparse** rainfall, and most central districts drifted into the **very light** to **light** categories. In the Jammu plains—particularly Jammu, Rajauri, and Mirpur—**light** to **moderate** rains persisted from late-season convective activity, though nowhere near the intensities seen in September. This rapid tapering of precipitation corresponded with the gradual recession of floodwaters outside the main Jhelum corridor.

Table 1. District wise average monthly rainfall

DISTRICT	AUGUST (in mm)	SEPTEMBER (in mm)	OCTOBER (in mm)
JAMMU	304.00	564.39	37.02
SAMBA	303.23	433.64	29.07
KISHTWAR	247.36	267.23	34.34
KATHUA	224.63	269.48	30.23
REASI	224.14	595.86	45.53
ANANTNAG	219.18	352.04	39.95
RAMBAN	209.97	540.98	40.19
RAJAURI	207.22	555.92	43.44
UDHAMPUR	206.81	500.26	36.23
DODA	185.83	374.10	33.14
MIRPUR	181.63	452.38	42.03
PULWAMA	177.16	268.36	37.88
KULGAM	173.47	450.42	40.29
SHOPIYAN	167.31	362.23	39.41
GANDERBAL	162.05	218.70	37.70
SRINAGAR	157.09	216.80	35.74
BADGAM	151.01	260.80	38.07
BANDIPURA	138.54	208.65	37.63
PONCH	121.90	288.15	41.11
BARAMOLA	110.69	212.00	38.47
MUZAFFARABAD	87.92	190.38	38.16
KUPWARA	87.66	119.72	37.55
TOTAL	4048.795	7702.473	833.1923

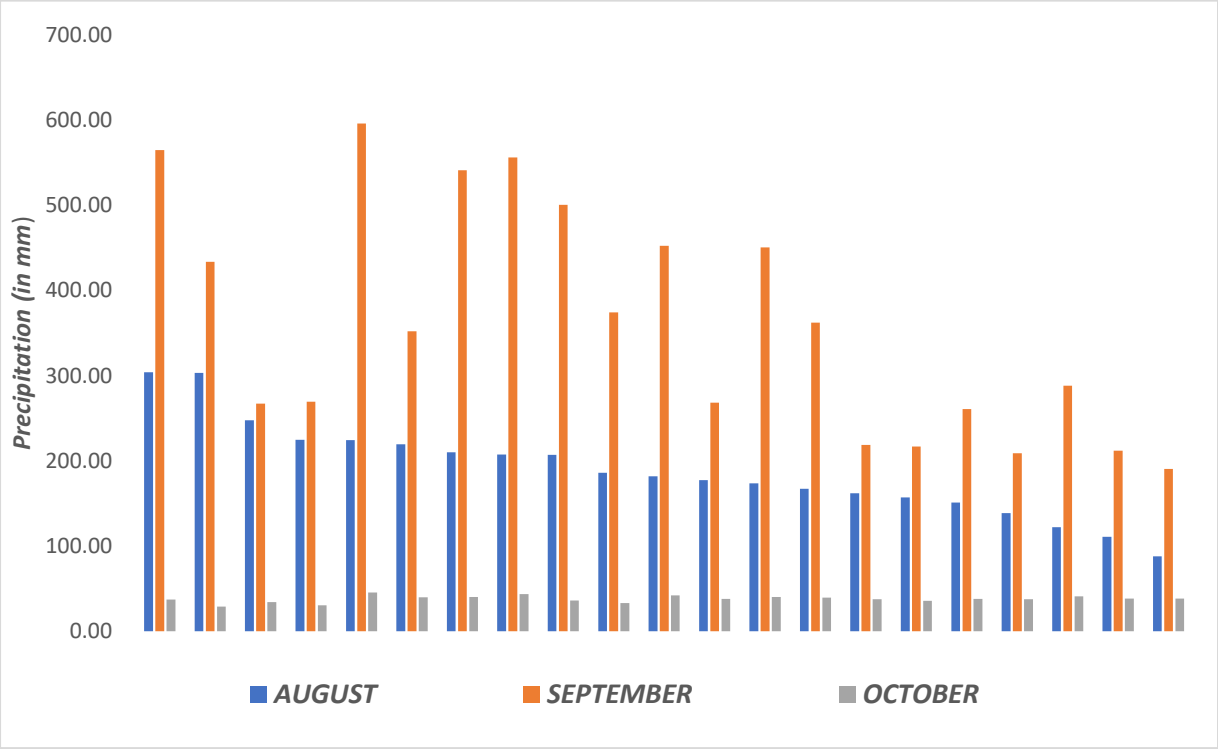


Chart 1. Comparison of rainfall across districts

3.2. Topographical Analysis

Jammu & Kashmir's terrain is a study in extremes, rising from roughly 200 m in the Jammu plains to over 6 600 m in the Greater Himalaya, with the Kashmir Valley (~400–1 600 m) nestled between two great mountain arcs.

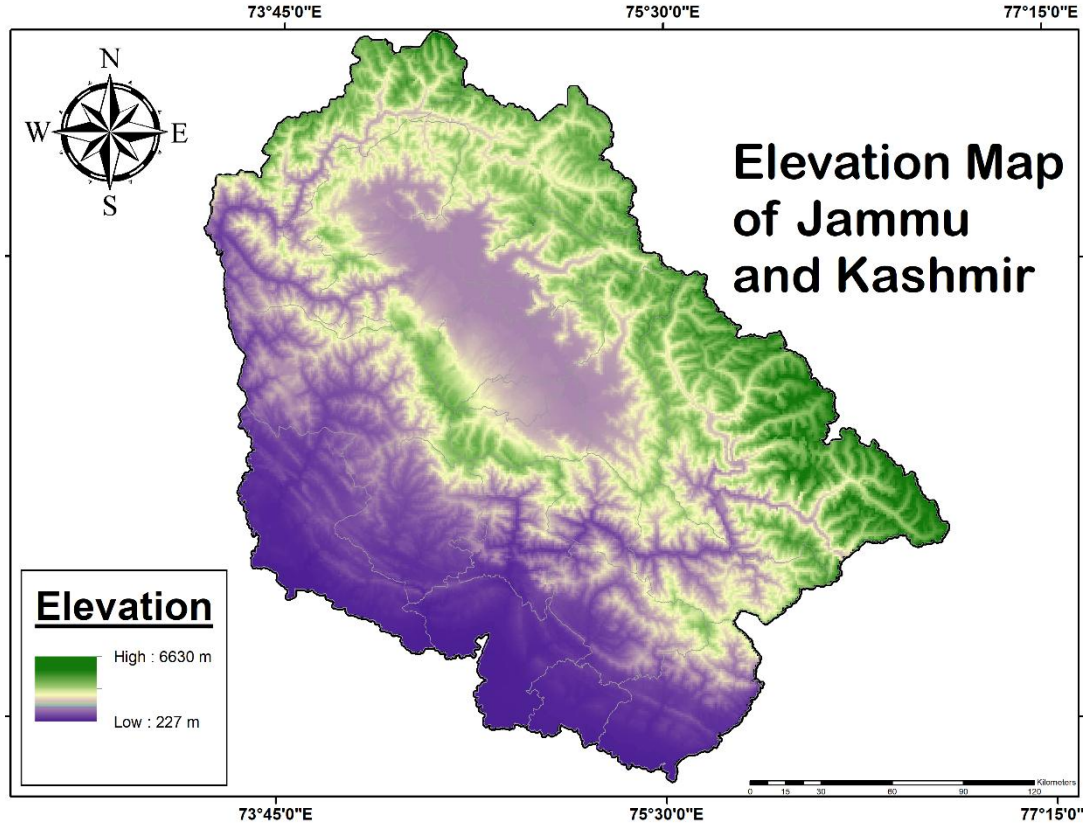
The Elevation Map shows a low-lying alluvial plain running north–south along the Jhelum River, bounded to the west by the Pir Panjal Range (peaks up to ~3 500 m) and to the east by the Zanskar and Greater Himalayan ranges (exceeding 4 400 m). This elevational zonation is reinforced by the Contour Map, where tight contour spacing on both flanks of the valley attests to very steep terrain, contrasted with broad, widely spaced contours in the central basin.

The Slope Map quantifies this, with the valley floor exhibiting slopes below 10° while the surrounding mountains present slopes often above 30°–40°, especially on the southern Pir Panjal escarpment.

The Aspect Map further reveals that slopes on the valley's western flank tend to face northeast while those on the east face southwest creating a checkerboard of sun-shadow exposures that influence snow-melt and runoff timing.
Finally, the Hillshade dramatizes the ridges and valleys, highlighting how deeply incised tributary channels carve the mountainside.

During the catastrophic September 2014 floods, this rugged topography played a central role. Torrential monsoon rains fell most heavily on the steep southern slopes of the Pir Panjal and the headwaters of the Lidder and Sind rivers. Those steep gradients (30°–40°) produced extremely rapid runoff, funnelling vast volumes of water through narrow mountain valleys. As these torrents crested, they poured onto the gentle, low-gradient floor of the Kashmir Valley, where the Jhelum River and its tributaries were unable to convey them downstream quickly enough.

The valley's broad, flat contours and low slopes (<10°) therefore became inundation zones, trapping water for days. Moreover, the orientation of many tributary outlets—aligned with aspect-controlled drainage patterns—meant that certain sectors of Srinagar and Budgam bore the brunt of overbank flooding. In sum, the interplay of high relief, steep slopes, and contrasting aspect-driven hydrology amplified runoff rates and created a natural bottleneck in the valley floor, transforming heavy rainfall into one of the worst flood disasters in the region's recorded history.

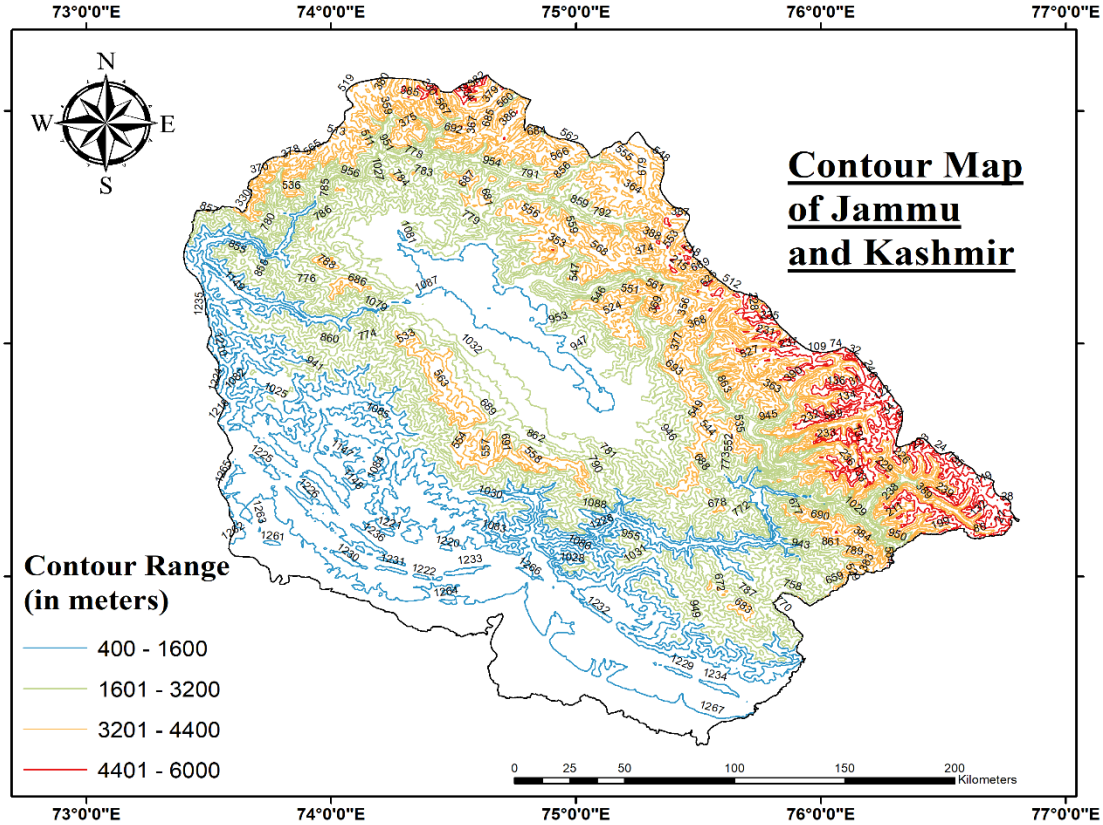


Map 7. Elevation map of Jammu and Kashmir

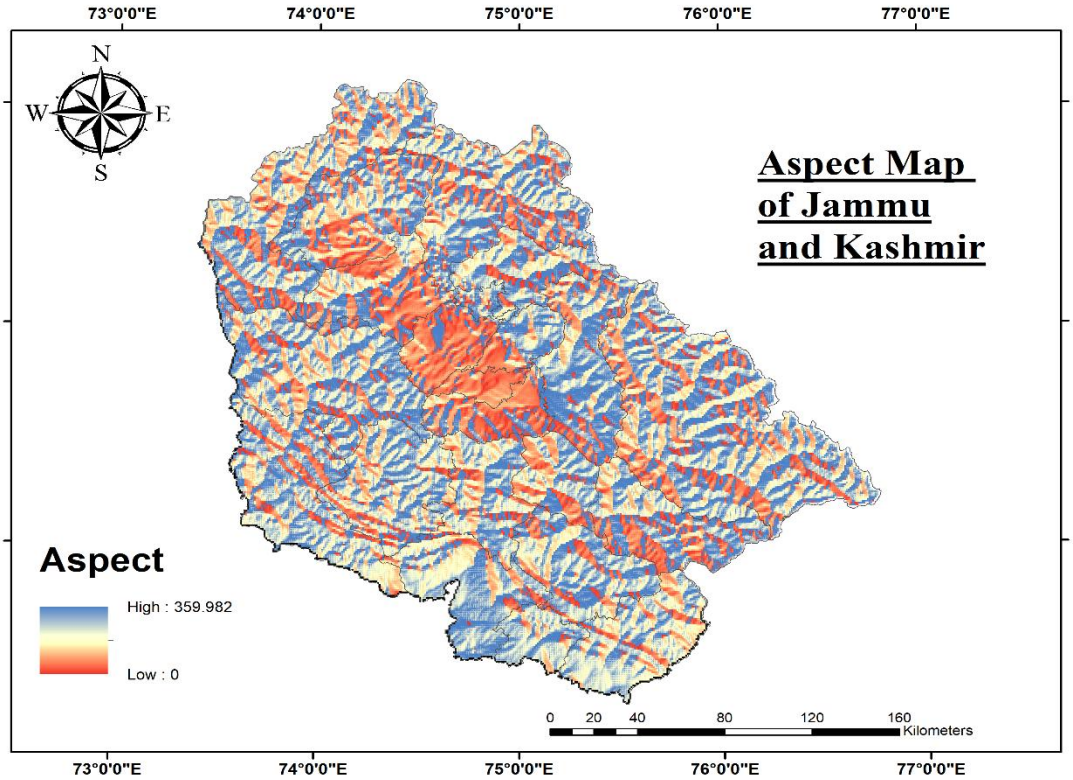
The elevation map of Jammu and Kashmir, revealing a complex topography varying from approximately 227 metres to 6630 metres. The southern and western parts of the region have low elevation ranging from 297 to around 1500 metres (low-elevation zones). This is the area which was the most affected due to extreme rainfall including regions of- Jammu, Samba, Kathua, Reasi, Rajouri, Doda. These regions were more conducive to water retention and accumulation.

The central region has moderate elevation. These regions likely experienced significant runoff rather than water retention, as steeper slopes would facilitate water flow downhill towards lower regions. Although rainfall in these areas was also moderate to heavy, the elevation suggests that water would have drained rapidly into lower areas, possibly increasing flood risks downstream.

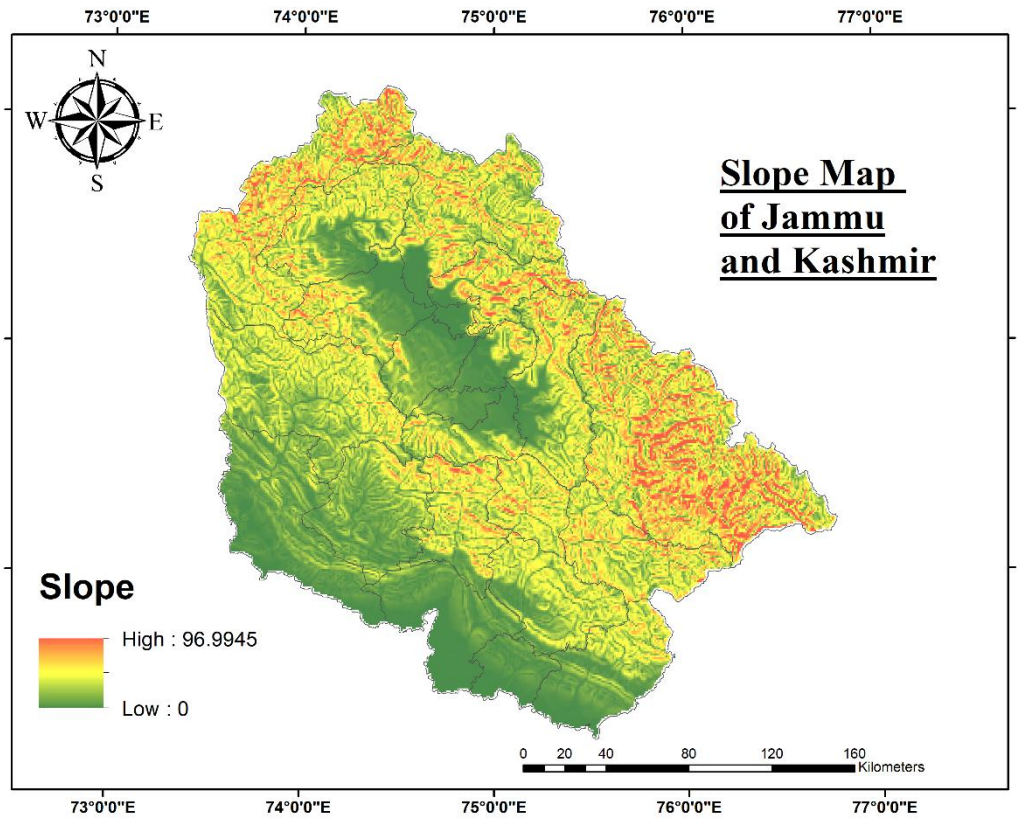
The northern and eastern regions exhibit high elevations of over 3,000 meters. These mountainous zones are less likely to retain water, with steep slopes causing rapid runoff towards the valleys and lower regions. These areas experienced light to moderate rainfall in September 2014, so despite their elevation, they were less affected by the floods. The steep slopes in these regions likely helped reduce standing water accumulation.



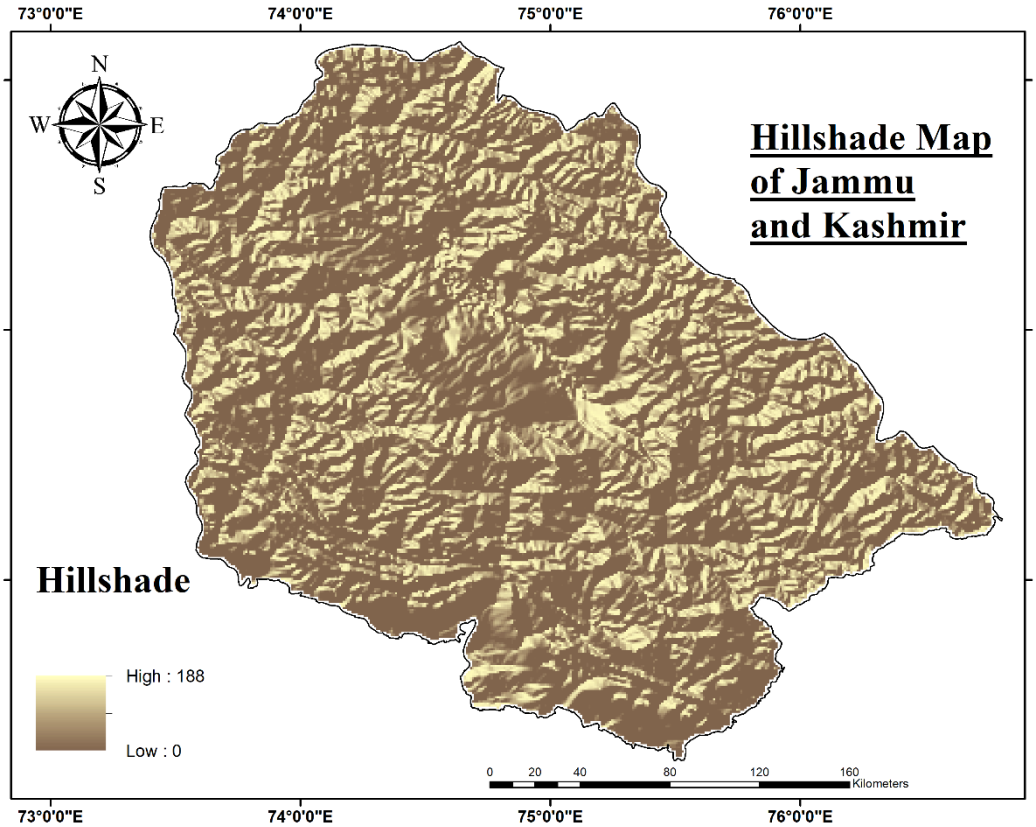
Map. 6. Contour map of Jammu and Kashmir



Map 5. Aspect map for Jammu and Kashmir



Map 8. Slope map of Jammu and Kashmir



Map 9. Hillshade map of Jammu and Kashmir

3.3. NDWI analysis

The Normalized Difference Water Index (NDWI) analysis for Jammu and Kashmir during the 2014 floods, comparing August (pre-flood) and September (post-flood) values, shows a notable increase in water presence across most districts. The data indicates that districts such as Ramban, Rajauri, Jammu, and Samba experienced the most significant increase in NDWI values, suggesting substantial inundation and surface water accumulation post-flood.

Jammu's NDWI nearly doubled from 0.055 to 0.122, and Samba, which initially had the lowest NDWI at 0.0245, rose sharply to 0.087, indicating extensive waterlogging in previously drier areas. Ramban recorded the highest post-flood NDWI (0.189), reflecting its vulnerability to water accumulation.

Conversely, Mirpur is the only district showing a decrease in NDWI, dropping from 0.164 to 0.124, possibly due to efficient drainage or water receding post-initial inundation. The spatial maps corroborate these changes, with darker blue areas (indicative of high NDWI) expanding notably in September, especially in the central and southern parts of the region.

Overall, the analysis highlights a widespread increase in surface water due to the floods, with spatial and quantitative data aligning to illustrate the severity and distribution of the flooding impact.

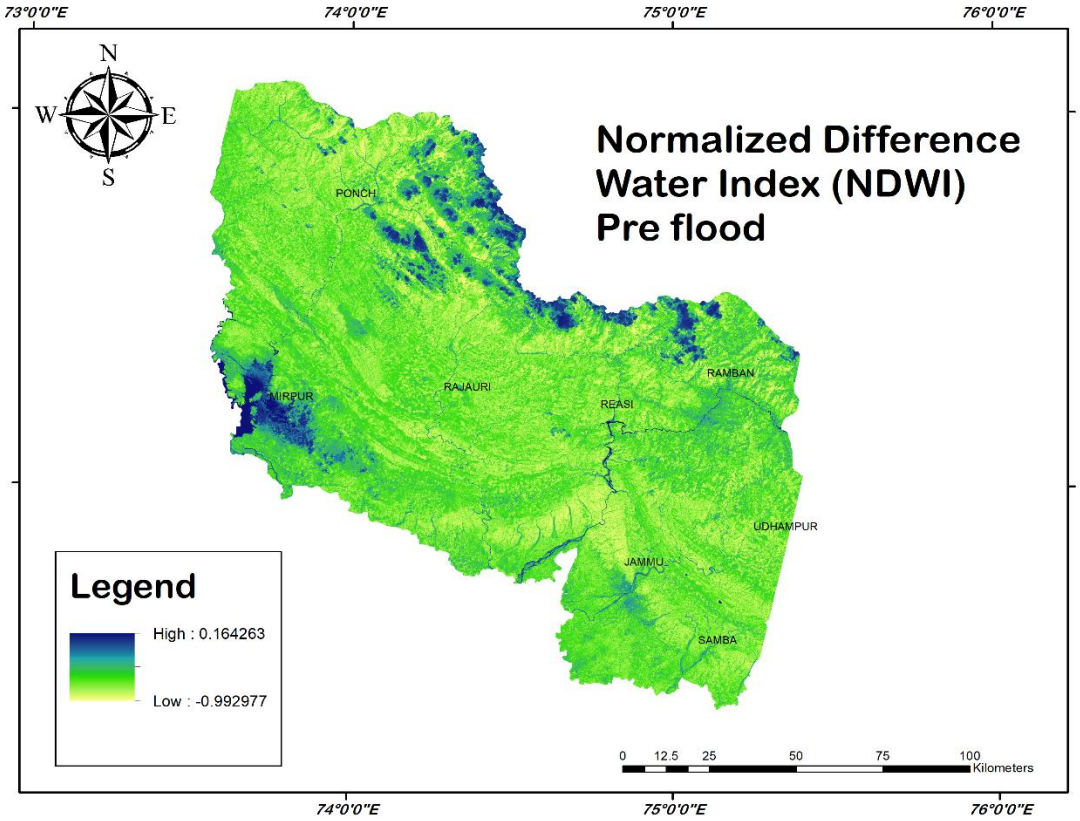
Some of the key inferences are:

- **Flood aftermath**: The decrease in NDWI values across the region suggests that the immediate aftermath of the flood was characterized by a reduction in surface water, possibly due to rapid drainage or water absorption into the ground.

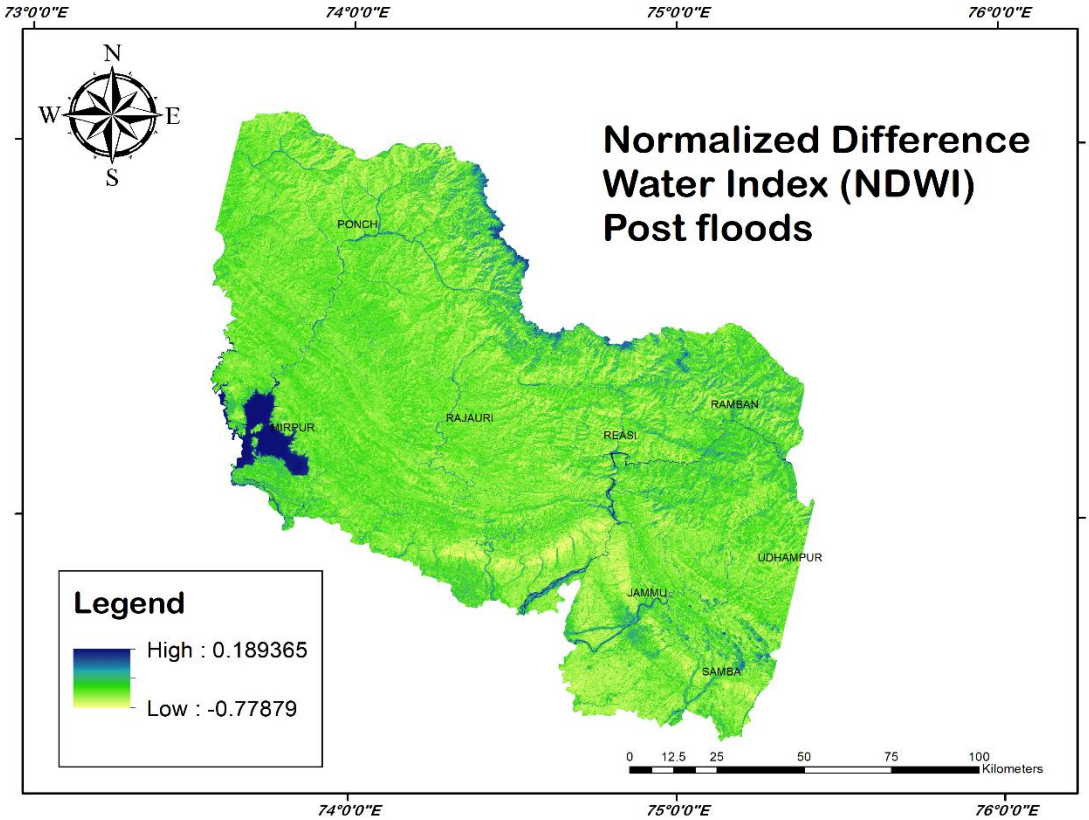
- **Changes in water distribution**: The less pronounced water patterns in the post-flood map indicate significant changes in water distribution, potentially due to alterations in river courses, creation of new drainage channels, or changes in land cover.

- **Soil moisture changes**: The general decrease in NDWI values could indicate lower soil moisture content post-flood, possibly due to increased runoff or changes in land surface characteristics.

The pre-flood data has cloud cover despite clipping and filtering, thus the image shows clouds as water. This has led to errors in the image, but the inferences have been made across different data sources.



Map 9. NDWI before flood (August 2014)



Map 10. NDWI after flood (September 2014)

Table 2. NDWI values for the chosen districts before and after the flood

DISTRICT	BEFORE	AFTER
MIRPUR	0.164263	0.124047
RAMBAN	0.158934	0.189365
REASI	0.104686	0.115841
PONCH	0.102544	0.127093
UDHAMPUR	0.090444	0.105043
RAJAURI	0.082341	0.110872
JAMMU	0.055069	0.122036
SAMBA	0.0245	0.087371

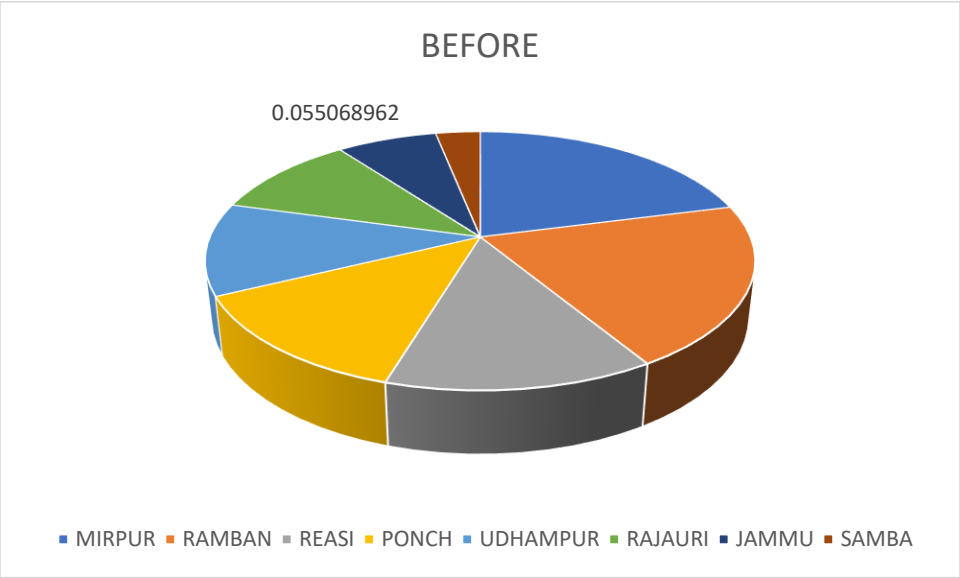


Chart 2. Pie Chart representing the NDWI proportions before flood

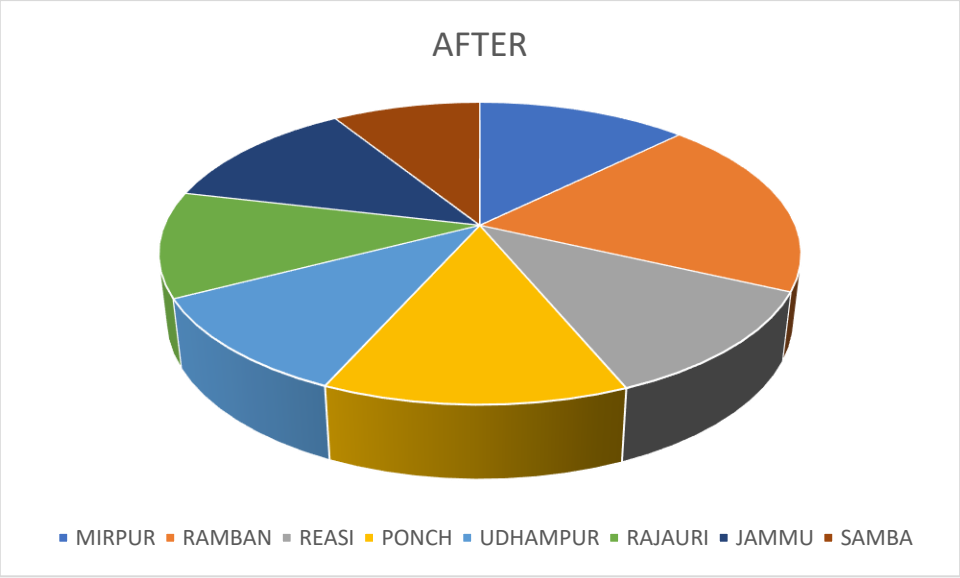
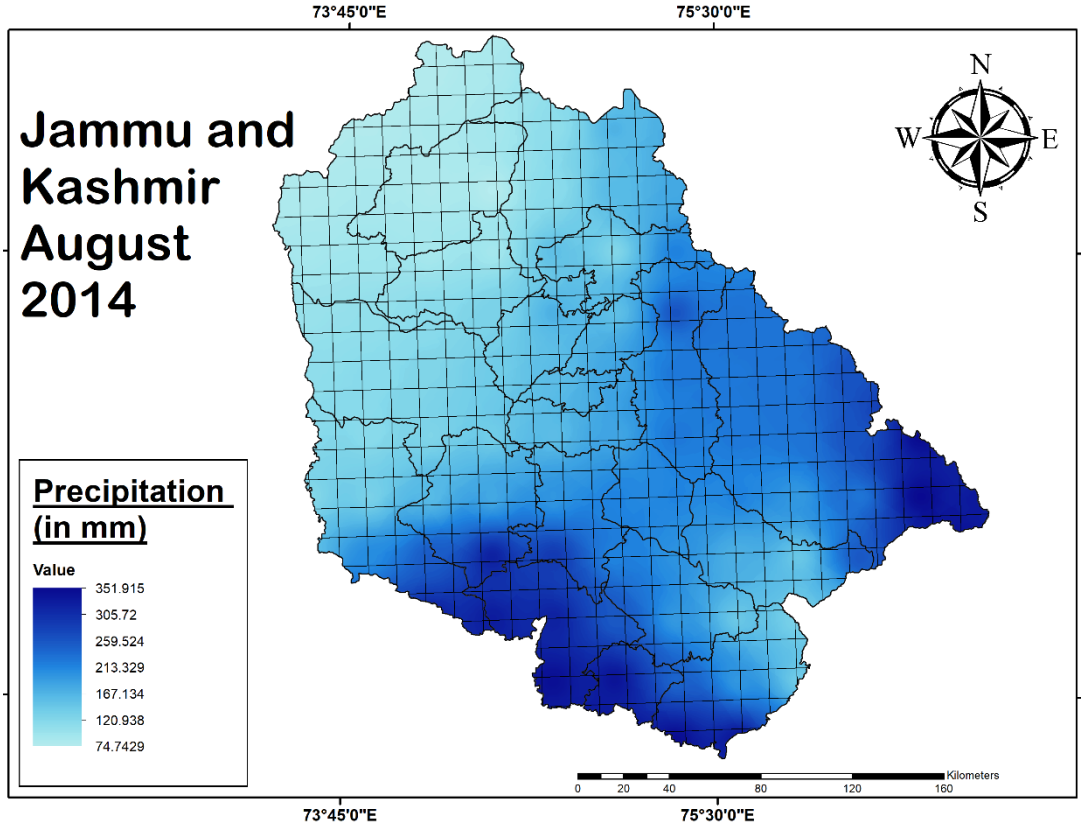


Chart 3. Pie Chart representing the NDWI proportions after flood

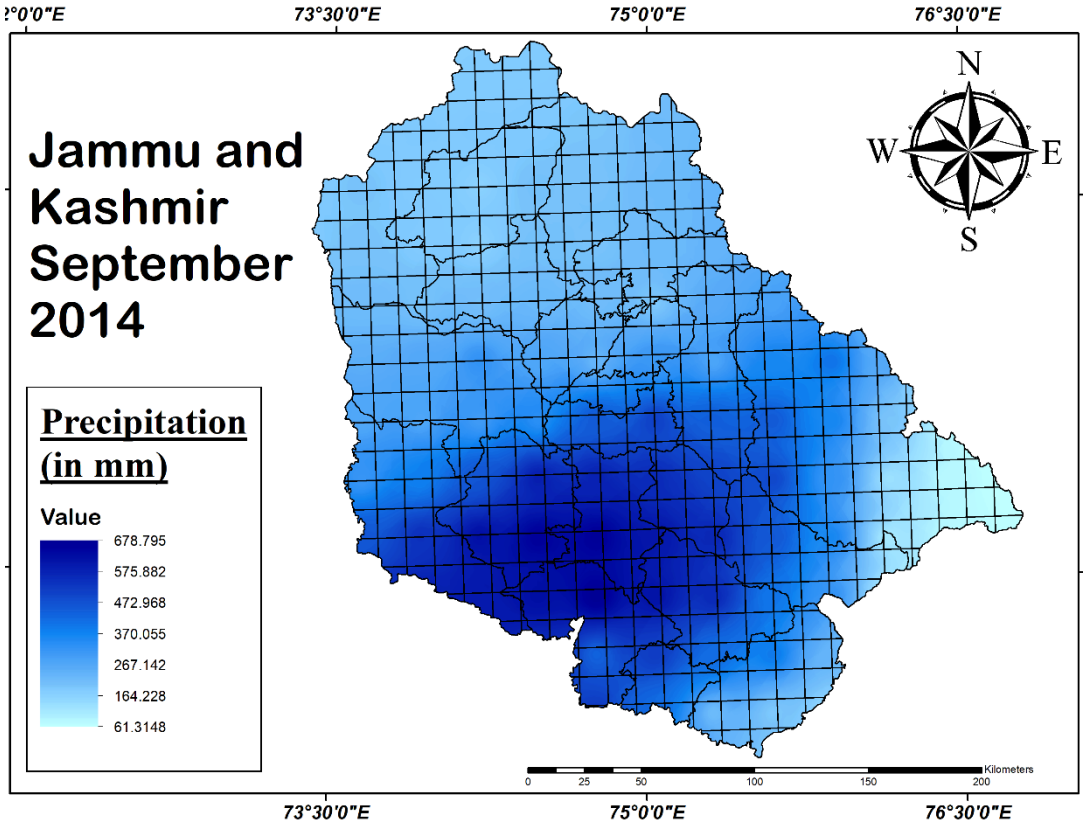
3.4. Fishnet analysis

Fishnet was created over the area of study for all the above data and analysed based on the gridded data in each cell. This gave a more comprehensive look at the changes in rainfall and flooding across time periods.

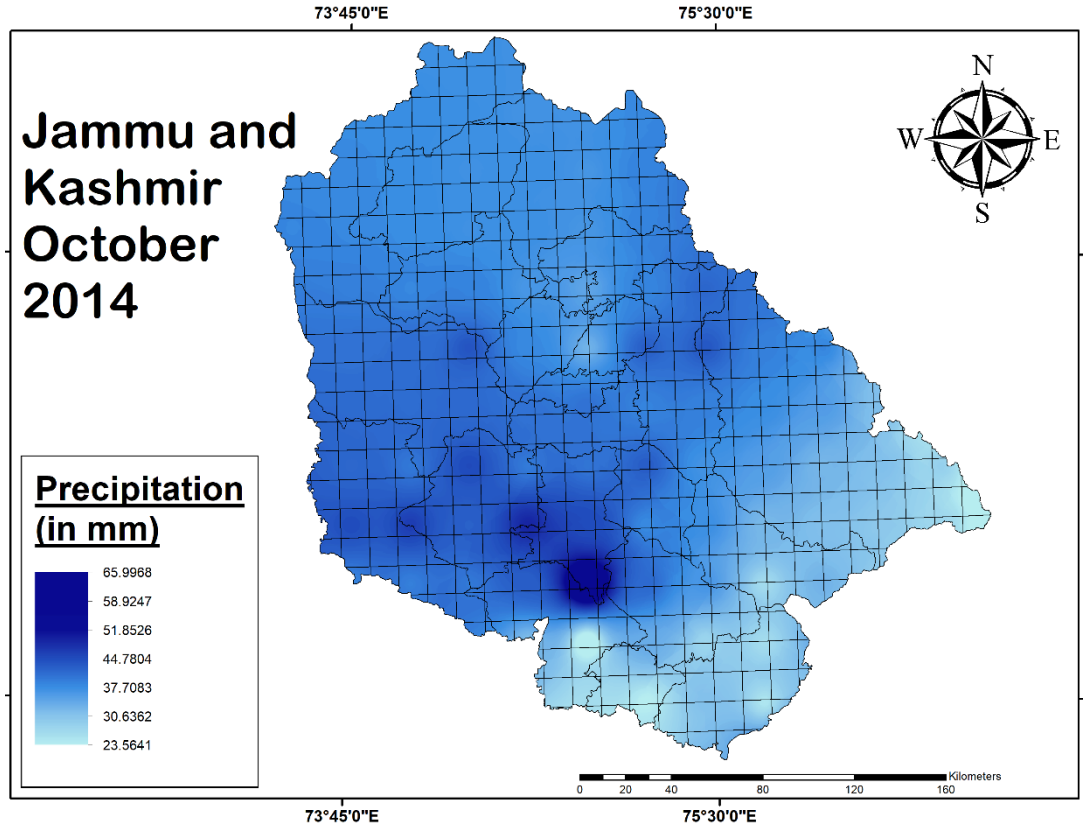
The respective maps of Rainfall have been added below that have been displayed in the gridded data as well. All other factors have been displayed in zonal statistics that is in Excel sheet attached with this report.



Map 11. Fishnet analysis of precipitation in August (2014)



Map 12. Fishnet analysis of precipitation in September (2014)



Map 13. Fishnet analysis of precipitation in October (2014)

Chart 3. Grid based zonal statistics

OBJECTID	AUG RAINFALL (in mm)	SEPT RAINFALL (in mm)	OCT RAINFALL (in mm)	NDWI_PRE	NDWI_POST	ELEVATION (in m)	ASPECT (in Degrees)	SLOPE (%)
1	344.99	249.13	29.81	-0.1665	-0.0422	282.02	177.00	0.47
2	324.61	230.96	31.56	-0.1332	-0.0945	292.32	200.20	0.63
3	323.17	205.54	33.05	-0.0517	0.0143	322.12	206.34	1.24
4	309.50	194.01	34.11	-0.0757	-0.0074	360.00	175.53	2.33
5	330.86	521.09	28.17	-0.0281	-0.0223	263.67	231.16	0.22
6	328.72	444.90	27.84	-0.1615	-0.1387	286.89	213.09	0.26
7	328.95	424.97	27.59	-0.0137	0.0528	302.41	175.62	0.78
8	315.25	345.79	27.00	-0.0668	-0.0312	311.36	198.29	1.35
9	309.57	247.54	28.87	-0.4028	-0.0101	335.96	207.43	2.81
10	293.30	239.32	30.92	-0.3955	0.0035	404.49	196.78	4.69
11	277.92	206.27	31.52	-0.4206	0.0134	575.89	139.65	5.64
12	252.27	194.61	29.96	-0.4003	-0.0366	554.54	187.17	2.86
13	207.65	173.79	29.61	-0.4167	0.0209	522.02	234.97	0.09
14	337.68	499.13	30.85	-0.4119	0.0403	262.36	233.86	0.13
15	337.52	496.57	29.25	-0.3803	-0.1308	266.38	247.03	0.34
16	327.03	469.24	28.36	-0.4237	0.0874	287.16	218.61	1.11
17	330.60	444.86	28.45	-0.4352	-0.0311	336.66	218.63	2.72
18	300.31	399.62	27.76	-0.4068	-0.0915	408.77	198.58	3.18
19	258.72	334.68	29.15	-0.4080	-0.0019	501.45	168.81	6.11
20	247.96	310.27	30.79	-0.4280	0.0839	613.14	199.49	6.62
21	214.98	284.49	30.50	-0.4082	0.0157	699.75	185.17	10.30
22	200.49	251.75	29.13	-0.4083	0.0210	879.03	151.75	12.11
23	164.93	210.25	3.71	-0.4304	0.0405	912.53	34.22	22.18
24	143.34	211.37	1.05	-0.3955	-0.1853	833.88	131.44	0.25
25	330.33	497.72	1.03	-0.4270	0.0799	257.03	356.99	0.24
26	329.72	460.51	4.32	-0.4161	-0.0254	267.30	241.32	1.75
27	315.35	482.46	5.27	-0.4199	0.0605	343.38	349.52	2.82
28	309.12	504.00	6.66	-0.4249	-0.0528	466.94	348.04	3.97
29	277.69	468.32	4.36	-0.4253	-0.0196	536.51	358.67	9.62
30	244.92	417.77	4.13	-0.4270	0.0847	818.04	339.84	11.73
31	222.99	391.75	2.14	-0.4121	0.1055	1087.26	350.60	16.05
32	197.70	371.04	1.50	-0.4063	-0.0943	1298.33	350.70	18.83
33	177.85	331.55	1.88	-0.4380	0.1064	1674.75	357.38	20.57
34	155.93	249.26	3.42	-0.4238	0.0358	1831.83	346.22	21.86
35	146.68	230.16	2.23	-0.4271	-0.0190	1736.39	239.82	0.47
36	318.03	586.74	1.55	-0.4229	-0.0570	277.59	125.31	0.32
37	323.06	580.00	6.35	-0.4264	0.0555	257.71	97.10	0.26
38	315.21	567.73	5.62	-0.3924	-0.0672	263.12	88.59	0.68
39	316.77	524.58	9.96	-0.4409	-0.0093	286.30	357.55	3.85
40	294.70	538.60	14.47	-0.4341	-0.0238	410.13	358.72	4.13
41	267.69	538.78	11.61	-0.4404	-0.0178	488.07	355.96	9.00
42	249.17	513.11	4.40	-0.4110	0.0207	656.71	356.35	9.54
43	211.54	473.91	5.14	-0.4152	-0.0001	926.33	355.30	15.92
44	197.24	439.44	5.43	-0.3787	-0.0286	1576.75	358.82	18.18
45	164.74	394.25	3.26	-0.4288	0.0830	2133.45	317.44	21.48

46	153.22	348.94	3.00	-0.4383	-0.0503	2980.49	314.63	21.98
47	142.99	290.69	3.26	-0.4279	0.0046	2460.56	148.04	23.02
48	136.35	244.79	1.95	-0.4288	0.1220	2915.79	56.93	1.41
49	290.86	590.87	0.68	-0.3835	-0.1010	336.95	265.67	1.89
50	304.09	599.24	0.83	-0.4266	0.0962	366.35	267.14	2.19
51	309.20	601.39	1.34	-0.4502	0.0705	362.88	356.03	2.53
52	316.57	612.68	5.18	-0.3870	-0.1011	338.91	329.79	2.74
53	308.16	613.25	6.42	-0.4512	0.0856	381.27	222.29	2.20
54	306.95	646.17	16.51	-0.4443	0.0367	361.11	354.06	4.98
55	281.38	616.18	27.21	-0.4103	-0.2269	560.39	330.86	5.66
56	255.49	600.19	22.45	-0.4525	-0.0363	682.23	327.83	5.82
57	235.18	571.53	6.98	-0.4100	0.0916	718.52	320.95	16.01
58	207.80	544.85	6.83	-0.4184	0.0361	1427.90	357.33	18.16
59	191.96	505.10	4.62	-0.4090	-0.0233	1976.32	350.52	19.95
60	168.65	415.51	5.27	-0.4186	0.0124	2778.15	358.55	19.95
61	158.75	378.46	3.77	-0.4347	0.0783	2995.17	354.75	19.17
62	153.33	329.59	3.03	-0.4155	0.0980	2770.02	359.47	21.86
63	155.09	282.39	1.65	-0.4532	0.0841	2706.55	351.40	25.79
64	178.22	183.44	1.42	-0.4641	-0.0320	3302.03	131.92	0.38
65	214.73	539.61	2.21	-0.4258	-0.1328	235.44	300.87	1.86
66	227.92	564.41	1.64	-0.4389	0.0750	314.40	352.14	3.93
67	257.98	580.11	1.93	-0.4483	0.0530	417.90	350.98	5.53
68	270.56	596.56	1.33	-0.4438	-0.0035	560.74	358.72	8.75
69	290.03	607.45	2.29	-0.4541	0.0366	590.69	356.46	7.64
70	306.06	620.53	3.03	-0.4561	0.0276	595.06	359.46	5.71
71	293.12	628.95	2.68	-0.4031	-0.0189	637.11	357.90	4.91
72	287.84	644.92	11.82	-0.4195	-0.0860	534.01	358.60	8.83
73	264.76	627.91	16.56	-0.4216	0.0072	662.85	359.63	18.67
74	232.05	603.93	18.11	-0.4304	-0.0475	1463.43	358.02	17.37
75	221.58	580.39	3.98	-0.3724	-0.0954	1433.56	355.27	20.10
76	207.31	551.14	4.15	-0.4150	-0.0642	1610.89	352.43	21.16
77	195.48	512.80	2.63	-0.3949	-0.1726	2049.96	359.87	23.56
78	177.96	450.30	4.42	-0.4243	0.0538	2436.67	358.53	16.54
79	170.84	402.15	5.26	-0.4436	0.0656	1884.59	356.16	17.14
80	159.47	347.91	3.45	-0.4492	-0.0049	2454.13	358.67	24.14
81	157.94	285.91	1.44	-0.4626	0.0764	2283.88	354.81	24.02
82	209.80	185.36	1.60	-0.4286	-0.0765	3122.91	359.95	25.89
83	252.58	124.47	1.27	-0.4565	-0.0793	4011.13	63.16	38.26
84	264.78	109.51	0.24	-0.4311	-0.0539	4330.72	292.45	2.11
85	169.40	468.54	2.02	-0.4219	-0.0042	292.33	334.93	1.66
86	197.72	505.20	3.94	-0.4326	0.0386	309.15	356.95	4.56
87	208.99	540.76	3.69	-0.4522	0.0282	422.21	353.99	7.37
88	232.53	564.27	4.67	-0.4526	0.0580	728.21	357.92	6.05
89	248.72	604.74	3.24	-0.4666	-0.0259	707.82	353.79	5.50
90	268.05	613.34	2.10	-0.4200	-0.0201	605.48	358.24	4.68
91	295.79	641.66	4.39	-0.4624	0.0802	691.73	358.93	7.36

92	272.15	639.18	5.10	-0.4671	0.0610	806.46	355.85	16.88
93	273.36	650.30	4.55	-0.5121	-0.0225	916.39	355.36	16.67
94	248.10	632.90	6.71	-0.4057	0.0487	858.86	359.39	22.16
95	222.49	605.75	8.70	-0.4033	-0.0149	1289.47	359.64	21.14
96	215.93	586.77	5.68	-0.4137	-0.0475	1413.91	358.07	19.12
97	206.84	560.50	2.13	-0.3830	-0.0552	1911.89	358.36	24.43
98	199.96	533.11	2.56	-0.4357	-0.1501	1574.64	359.05	24.63
99	187.37	470.64	3.99	-0.4183	-0.0502	1636.38	358.74	24.32
100	182.77	435.35	2.97	-0.4046	0.0363	1649.00	357.62	28.09
101	176.64	365.95	0.88	-0.3059	-0.0774	1810.51	355.11	28.80
102	178.33	323.90	0.87	-0.4328	-0.2251	2446.95	352.17	25.82
103	216.91	187.14	0.61	-0.4496	0.0491	3720.56	359.90	30.75
104	249.64	120.88	0.92	-0.4338	-0.0479	3701.97	350.93	39.69
105	283.40	104.39	0.88	-0.4495	0.0914	3517.82	355.76	32.75
106	326.65	80.14	1.04	-0.4440	0.1040	4285.82	350.91	33.97
107	336.37	72.24	1.73	-0.4351	-0.0282	4644.54	353.71	24.95
108	333.85	67.86	2.56	-0.4551	-0.0850	4844.13	358.28	1.81
109	151.89	442.28	0.93	-0.4444	-0.0388	364.99	359.77	2.03
110	164.08	464.03	1.22	-0.4283	-0.0298	385.17	348.92	6.55
111	171.83	506.74	1.90	-0.4492	-0.0823	662.12	353.68	4.99
112	189.65	525.46	2.52	-0.4665	-0.0502	849.49	346.48	7.59
113	198.02	571.35	2.31	-0.4697	0.0216	841.41	356.69	5.04
114	213.87	581.45	0.73	-0.4605	0.0150	828.02	335.81	7.27
115	217.31	613.91	4.04	-0.3691	-0.0418	920.81	353.73	11.82
116	223.98	615.22	4.71	-0.4485	0.1006	1302.09	280.20	19.24
117	219.40	622.04	4.00	-0.5118	0.0816	1429.74	352.18	23.81
118	219.86	610.40	2.31	-0.5322	0.0725	1382.88	356.98	22.41
119	216.74	588.77	3.42	-0.5473	-0.0857	1962.29	356.69	23.53
120	213.49	569.20	3.87	-0.3648	-0.0287	1332.63	358.82	23.62
121	211.75	543.56	2.45	-0.4199	-0.2443	1658.32	357.78	21.42
122	208.36	515.49	1.47	-0.4517	-0.0838	2535.62	344.35	22.05
123	206.63	477.10	2.80	-0.3962	-0.1027	2683.44	358.30	21.91
124	205.95	436.16	3.41	-0.4506	-0.0560	2725.15	352.61	31.40
125	198.40	368.23	2.10	-0.4461	0.0123	2227.82	358.60	33.09
126	199.41	304.05	1.89	-0.4189	0.0680	3250.60	358.80	35.97
127	223.76	210.14	1.20	-0.4482	-0.1625	3146.03	355.72	37.95
128	237.58	141.17	1.45	-0.4495	-0.0025	3004.30	352.44	29.97
129	286.96	109.11	1.21	-0.4421	-0.0613	4475.97	358.15	28.43
130	340.67	81.85	1.49	-0.4138	-0.0959	4542.25	355.45	30.09
131	333.56	72.02	2.41	-0.4375	-0.1875	4663.81	340.02	17.31
132	330.90	64.96	2.63	-0.4164	0.0849	5118.59	39.53	3.56
133	123.26	338.06	0.29	-0.4362	0.0861	423.50	336.62	2.16
134	129.54	357.65	1.53	-0.4154	0.0989	403.57	347.26	4.92
135	137.04	392.29	1.91	-0.4413	0.1066	522.74	356.86	7.52
136	138.74	405.37	2.69	-0.4456	0.0117	827.80	356.31	6.81
137	157.98	448.08	3.93	-0.4366	-0.0336	896.35	351.25	9.44

138	164.54	481.92	2.64	-0.4549	-0.0244	1135.59	356.48	6.42
139	177.81	516.47	1.58	-0.4522	-0.0282	976.80	352.14	10.58
140	182.81	569.20	2.73	-0.4348	-0.0216	1399.35	356.99	17.03
141	191.78	572.79	5.75	-0.4624	-0.0986	1817.61	359.73	19.81
142	196.57	575.86	5.05	-0.4321	-0.0059	1744.32	347.96	23.71
143	200.99	571.39	3.87	-0.4304	-0.0390	1739.79	356.33	24.74
144	208.24	557.62	2.99	-0.3923	-0.0572	2865.44	303.79	24.51
145	209.84	545.07	2.43	-0.4082	0.0911	2402.17	353.50	22.01
146	211.93	515.31	4.20	-0.5384	0.1168	2160.57	357.77	22.04
147	213.58	498.38	2.11	-0.5065	0.0805	3203.10	353.49	21.75
148	213.56	481.88	1.57	-0.4679	-0.0188	3073.55	359.60	23.29
149	213.61	460.09	2.15	-0.4417	-0.1104	2020.37	358.31	33.97
150	209.68	362.38	2.35	-0.4542	-0.0948	2227.54	358.29	34.97
151	211.38	310.50	2.10	-0.4494	0.0479	2810.81	274.23	36.80
152	233.35	210.40	0.85	-0.4335	0.0269	3781.55	338.62	34.74
153	250.36	140.87	1.24	-0.4391	0.0278	4170.76	357.47	37.91
154	291.11	113.49	1.41	-0.4433	-0.0224	3836.05	359.74	33.72
155	329.84	80.69	1.71	-0.4216	-0.0866	4182.24	356.66	21.46
156	328.45	71.58	3.03	-0.4794	-0.1244	4989.10	57.95	17.15
157	324.02	64.64	1.19	-0.4446	-0.0381	5371.22	63.92	3.86
158	126.56	329.81	0.14	-0.4446	0.1119	465.83	357.86	4.47
159	126.52	321.18	0.89	-0.4263	0.1049	515.52	357.52	7.66
160	129.30	344.15	0.55	-0.4535	-0.0740	648.50	357.60	8.07
161	131.67	365.50	1.37	-0.4282	0.1158	758.72	356.03	11.63
162	140.17	392.31	1.50	-0.4116	0.1050	1086.11	352.65	11.71
163	146.68	434.34	3.09	-0.4194	-0.0321	1428.62	328.31	10.56
164	155.73	456.41	2.21	-0.4530	0.1099	1323.55	359.64	20.07
165	160.56	506.43	3.09	-0.4431	-0.0226	2093.16	357.47	20.07
166	169.92	510.25	2.13	-0.4443	0.0379	3373.07	342.93	26.33
167	174.14	524.08	1.75	-0.3935	0.1894	2993.77	355.73	23.83
168	182.49	524.42	1.46	-0.4526	0.1028	3440.63	358.09	25.00
169	191.28	521.76	1.09	-0.5522	0.0979	3308.00	358.89	22.20
170	198.96	513.05	2.48	-0.4732	0.0344	2674.63	358.28	19.83
171	214.60	488.71	3.85	-0.4254	-0.0778	2427.02	356.76	18.96
172	219.36	472.85	2.50	-0.4251	-0.1281	2720.15	349.68	24.36
173	219.99	459.66	1.83	-0.4423	-0.1512	3099.59	309.85	25.41
174	221.77	436.22	1.85	-0.4413	0.0065	2772.02	356.75	35.33
175	222.24	374.84	2.23	-0.4634	0.0599	2637.69	358.28	35.59
176	227.49	320.46	3.05	-0.4516	0.0520	3201.93	359.38	30.76
177	245.73	242.39	2.28	-0.4343	-0.0265	4288.11	355.39	27.19
178	263.58	174.37	2.17	-0.4345	-0.0370	4612.03	348.54	29.97
179	294.32	124.55	2.68	-0.4426	-0.0890	4345.99	272.51	26.95
180	334.83	84.58	2.65	-0.4438	-0.0515	4756.47	135.96	16.41
181	332.32	71.85	1.39	-0.4506	-0.0473	5276.40	351.88	6.95
182	120.19	294.24	0.69	-0.4440	0.1027	641.58	345.73	7.83
183	120.21	307.29	0.48	-0.4414	0.0212	776.83	340.30	9.88

184	119.69	306.28	0.64	-0.4642	0.0451	847.80	358.68	9.35
185	126.89	330.58	0.64	-0.2444	-0.0403	1151.13	356.81	11.68
186	128.30	336.73	1.49	-0.4408	-0.0223	1348.60	352.22	14.51
187	138.28	365.12	1.96	-0.4159	0.1560	1738.24	357.14	19.89
188	141.50	374.35	2.26	-0.4436	0.0801	2660.75	357.79	17.85
189	152.38	402.67	0.92	-0.4679	0.1640	3751.91	357.96	21.01
190	159.18	450.98	0.59	-0.4246	0.1551	3485.40	356.31	14.69
191	170.27	462.31	0.49	-0.3858	-0.0517	2688.59	358.43	9.37
192	182.90	485.81	0.86	-0.4737	0.0915	2094.92	358.47	8.92
193	193.54	485.01	1.83	-0.5547	0.1240	1918.14	358.12	7.29
194	215.26	450.66	1.76	-0.4582	0.1199	1908.47	352.96	15.12
195	222.68	440.65	1.19	-0.5947	-0.0530	2421.08	358.65	18.75
196	223.30	439.70	1.59	-0.4027	0.0047	3515.91	343.27	22.19
197	225.66	434.98	1.94	-0.4146	-0.1550	3434.78	357.03	34.60
198	227.72	376.91	2.18	-0.4340	-0.0928	2926.80	356.91	34.72
199	231.40	347.09	2.40	-0.4391	-0.1082	4031.54	356.98	38.11
200	248.75	256.64	2.55	-0.4342	0.0402	4425.94	357.70	25.90
201	266.11	193.86	2.88	-0.4447	0.0636	4735.97	270.70	22.84
202	294.00	136.32	2.73	-0.4469	-0.0804	5055.23	348.75	10.25
203	322.33	89.85	0.22	-0.4381	-0.0718	5477.00	357.77	10.91
204	117.37	278.21	0.14	-0.4301	-0.0759	975.21	355.72	12.72
205	116.53	274.50	0.62	-0.4487	-0.0620	1276.78	358.61	11.08
206	117.61	281.01	0.46	-0.4724	0.0471	1205.64	359.11	14.92
207	119.25	282.24	0.45	-0.4649	0.0373	1102.89	325.74	18.73
208	123.67	296.13	0.25	-0.4454	-0.0234	1480.71	319.44	22.57
209	128.11	308.82	0.54	-0.4683	-0.0551	2003.86	350.55	18.72
210	134.60	323.18	0.24	-0.4615	0.1405	3043.84	359.10	11.07
211	141.34	334.76	1.19	-0.4839	0.1467	3767.09	358.80	3.91
212	149.72	350.97	1.19	-0.3733	-0.0164	2650.08	116.42	2.07
213	158.21	386.60	1.66	-0.5155	-0.1034	2068.36	357.71	0.80
214	168.39	395.45	1.81	-0.4754	0.0936	1764.00	356.06	5.41
215	181.92	413.93	1.78	-0.4173	0.0885	1615.16	357.74	17.69
216	193.79	417.08	1.04	-0.5630	0.0838	1752.33	359.59	19.88
217	213.05	394.81	1.42	-0.4641	0.0001	2490.20	359.80	28.68
218	222.30	391.61	1.38	-0.5043	0.0331	3641.30	357.29	26.50
219	224.73	397.18	1.84	-0.4679	-0.0504	3160.36	358.95	32.02
220	226.68	395.71	1.96	-0.4206	-0.1299	3700.87	353.41	29.16
221	228.95	372.89	1.73	-0.4365	-0.1431	3836.46	358.03	24.42
222	231.51	346.34	1.97	-0.4428	0.0328	4646.77	124.67	31.45
223	248.68	279.65	2.67	-0.4445	0.0424	4861.61	353.68	16.77
224	264.55	219.68	2.76	-0.4518	-0.0440	5248.89	355.79	11.71
225	275.67	171.46	1.09	-0.4565	0.0486	1143.91	354.45	14.89
226	110.58	256.76	0.08	-0.4495	0.0124	1726.87	357.53	11.54
227	111.62	258.04	0.67	-0.4637	0.0456	1407.78	354.48	16.49
228	114.86	264.58	0.62	-0.4700	0.0438	1227.96	355.70	19.79
229	117.03	272.11	0.56	-0.5548	0.0325	1670.53	358.11	16.81

230	121.18	282.54	0.37	-0.4868	0.0042	2188.42	354.55	11.77
231	125.33	308.36	0.91	-0.8024	0.0120	3705.89	358.02	6.13
232	131.26	307.51	1.75	-0.4477	0.0194	3200.06	356.58	3.42
233	138.29	293.21	3.26	-0.4764	0.0174	2300.23	355.06	1.50
234	145.45	295.70	2.21	-0.4566	-0.0171	1863.70	351.87	4.60
235	155.96	279.26	2.89	-0.3942	-0.0393	1645.27	351.19	7.81
236	164.65	283.77	5.46	-0.4354	0.0584	1689.52	343.84	21.01
237	179.39	319.79	5.75	-0.5013	0.0706	1833.95	359.59	23.01
238	189.80	324.03	4.12	-0.5804	-0.0181	2863.25	349.27	26.78
239	214.40	316.02	2.74	-0.5107	0.0113	3541.53	359.50	25.25
240	224.43	307.79	3.18	-0.3879	0.0344	3727.64	356.47	26.40
241	225.68	350.17	3.88	-0.6084	-0.1232	3992.24	358.35	29.17
242	227.24	360.78	2.49	-0.6622	-0.0348	4330.88	345.88	27.88
243	229.65	365.43	1.36	-0.4371	-0.0026	4496.07	356.13	16.89
244	231.01	377.45	2.18	-0.4526	-0.1704	4866.03	359.46	13.95
245	246.09	294.71	4.28	-0.4617	-0.0877	785.00	357.54	17.78
246	259.78	263.58	3.11	-0.4533	-0.0052	1288.62	337.39	18.87
247	104.56	237.79	0.06	-0.4517	0.0087	1560.22	357.65	19.19
248	108.39	245.69	0.51	-0.3452	0.0164	1899.32	354.54	20.12
249	110.59	247.99	0.42	-0.5468	-0.0198	2096.23	350.14	16.02
250	112.81	256.47	0.65	-0.5471	0.0483	2168.12	356.02	11.89
251	116.47	261.38	0.64	-0.4901	-0.1061	2734.43	356.41	4.41
252	120.35	286.17	1.56	-0.5347	0.0814	3774.53	359.14	1.99
253	126.24	278.97	2.88	-0.9585	0.0666	2469.24	358.81	4.07
254	133.15	267.37	4.50	-0.6587	0.0367	1930.48	356.23	13.88
255	143.07	263.61	1.76	-0.5058	0.0771	1659.21	330.73	22.57
256	155.89	251.08	1.67	-0.4751	0.0263	1649.47	358.63	23.99
257	164.41	253.47	3.93	-0.5142	0.0377	1943.11	351.74	27.21
258	169.71	274.19	5.47	-0.4574	-0.0015	2611.59	356.89	23.98
259	183.21	281.87	4.84	-0.4602	-0.0221	3432.66	357.92	25.14
260	217.02	283.66	2.52	-0.3802	-0.0483	3617.87	357.74	22.99
261	234.34	292.26	2.80	-0.4273	0.0395	4182.16	357.70	23.13
262	229.27	315.70	2.55	-0.6257	0.0490	4355.91	298.28	20.81
263	229.65	331.34	2.09	-0.4451	-0.0326	4592.04	349.40	16.09
264	229.48	338.29	1.52	-0.4265	0.0339	4877.70	345.63	16.49
265	230.77	345.44	0.92	-0.4029	-0.0206	1067.98	359.29	20.54
266	239.80	290.60	2.36	-0.3685	-0.0617	1378.11	359.18	21.27
267	101.82	223.19	0.87	-0.7309	-0.0671	1621.08	359.75	21.63
268	102.69	224.98	0.83	-0.6497	-0.0628	2382.90	359.25	20.52
269	106.05	225.31	1.00	-0.5529	-0.0649	2287.65	359.73	15.32
270	108.07	219.46	1.28	-0.4369	-0.0533	2612.46	359.03	5.41
271	111.67	225.02	2.01	-0.6300	-0.2144	3318.26	358.73	1.93
272	116.36	226.97	2.20	-0.4698	-0.0367	2534.40	358.51	2.75
273	121.00	230.55	2.90	-0.4828	0.0246	1800.26	349.23	13.16
274	128.18	234.54	3.61	-0.4480	0.0211	1648.80	355.56	20.70
275	137.37	234.03	2.33	-0.4680	0.0977	1615.22	356.65	19.61

276	160.76	227.62	1.59	-0.3618	0.1080	1937.80	344.62	26.57
277	163.64	225.79	1.98	-0.5981	0.1109	2797.65	354.97	26.07
278	159.85	220.19	3.15	-0.5410	0.0687	3426.70	359.13	28.14
279	169.67	215.73	2.98	-0.4605	0.0406	3302.47	343.97	27.04
280	221.51	251.75	1.96	-0.5033	0.0461	4019.21	353.97	17.83
281	244.87	263.40	1.90	-0.9930	-0.0284	4006.62	350.32	14.56
282	228.13	284.63	0.83	-0.7723	-0.0068	4700.21	359.51	18.57
283	228.85	292.68	1.12	-0.4688	-0.0443	1168.10	358.83	23.14
284	228.36	308.59	1.29	-0.4463	-0.0151	1609.51	354.61	24.67
285	98.25	211.53	0.29	-0.5333	-0.0183	1764.55	354.71	23.73
286	98.79	213.64	0.52	-0.4090	-0.0252	1814.00	358.36	18.56
287	100.79	211.25	0.47	-0.4478	-0.1354	2159.42	359.63	5.62
288	102.66	204.08	0.90	-0.4215	-0.0055	2093.20	358.71	1.58
289	103.90	205.50	0.87	-0.4286	-0.0190	2192.10	354.66	0.20
290	106.10	207.14	0.92	-0.4286	0.0657	1834.81	329.04	5.67
291	110.39	210.31	0.79	-0.4271	0.0643	1626.17	358.69	26.19
292	115.57	218.72	1.39	-0.4166	0.0711	1581.49	358.78	23.28
293	126.45	219.12	1.36	-0.4211	0.0317	1670.29	358.19	25.01
294	145.07	214.33	0.70	-0.6709	-0.1161	2550.69	359.20	23.89
295	153.33	214.44	1.99	-0.6984	0.0069	3463.56	354.44	21.61
296	152.35	211.61	2.83	-0.5529	-0.0633	3682.27	356.24	28.50
297	163.68	214.34	1.66	-0.6740	-0.1667	4060.52	353.57	16.99
298	198.46	237.42	1.19	-0.6123	0.0374	4194.28	351.78	17.97
299	216.39	252.22	2.09	-0.5309	-0.0793	4298.57	351.93	22.07
300	216.09	265.67	1.14	-0.5196	0.0893	1129.74	359.15	23.77
301	218.82	282.66	0.80	-0.4233	-0.0445	1387.26	359.54	22.91
302	218.41	280.19	0.17	-0.6188	0.0159	2114.75	359.70	23.38
303	91.43	198.09	0.75	-0.3891	-0.0092	2531.20	354.61	10.79
304	91.97	200.85	0.68	-0.4346	-0.1600	3246.73	356.88	1.41
305	94.62	199.07	0.73	-0.4312	-0.0818	2918.96	354.66	0.19
306	95.58	191.49	0.93	-0.4200	0.0069	1985.57	353.16	11.88
307	95.25	190.59	0.88	-0.4353	-0.2051	1601.44	356.70	22.90
308	92.16	184.25	0.77	-0.4857	-0.0253	1579.07	355.17	26.62
309	97.75	187.82	1.28	-0.3576	-0.0811	1863.22	358.74	26.89
310	96.62	203.61	1.17	-0.4378	-0.2463	2260.38	358.80	28.62
311	110.97	206.44	0.77	-0.6162	0.0693	2622.46	357.08	29.96
312	142.79	204.06	0.97	-0.6193	0.0502	3346.43	353.26	29.79
313	147.79	204.59	1.06	-0.6575	0.0502	3425.37	352.30	17.08
314	137.78	214.73	1.62	-0.6209	0.0374	3702.53	357.05	22.98
315	141.58	220.64	1.57	-0.6726	0.0684	3826.29	358.38	24.50
316	189.42	230.70	1.09	-0.3905	0.1271	1080.93	359.59	21.35
317	210.50	237.66	1.52	-0.4028	-0.0058	2097.74	357.23	19.72
318	208.87	251.30	0.53	-0.4416	0.0092	1975.46	358.06	11.10
319	87.36	190.65	0.22	-0.5237	-0.0063	2351.38	358.40	3.13
320	88.45	195.81	0.36	-0.4973	-0.0676	3175.83	348.05	9.80
321	90.24	192.78	0.36	-0.4494	-0.0084	2110.13	341.57	6.18

322	91.77	184.86	0.48	-0.6509	0.0704	1666.31	358.90	15.24
323	90.24	182.72	0.47	-0.6036	0.0556	1830.62	356.14	24.18
324	89.48	175.88	0.46	-0.6760	0.0575	1708.58	343.49	23.12
325	91.86	179.31	0.67	-0.6084	0.0033	1939.88	358.20	18.79
326	93.92	196.36	0.34	-0.3708	0.1248	3270.59	358.21	20.15
327	105.12	201.83	0.19	-0.4085	0.0787	3637.71	353.24	20.16
328	125.16	200.97	0.42	-0.4574	-0.0227	3856.53	350.28	22.77
329	136.03	203.27	0.42	-0.5306	-0.0236	3911.65	357.38	24.96
330	144.30	212.44	0.96	-0.4664	0.0686	4264.64	354.21	28.09
331	157.17	219.59	1.33	-0.4783	-0.0215	1612.20	355.73	26.55
332	178.13	227.50	0.70	-0.6303	-0.0659	1689.08	358.36	16.73
333	191.73	234.45	0.84	-0.6080	-0.0004	2469.09	356.56	6.29
334	83.12	188.56	0.12	-0.5581	0.0463	2356.70	355.59	7.22
335	83.59	193.42	0.14	-0.4940	0.0187	2593.78	358.46	13.15
336	85.84	191.17	0.15	-0.4222	0.0874	1871.95	353.13	21.72
337	87.14	184.39	0.33	-0.5064	-0.1037	1722.91	355.30	24.27
338	86.09	181.70	0.25	-0.4580	-0.0748	1901.00	359.66	19.71
339	83.68	175.42	0.26	-0.6095	0.0036	2658.92	357.36	19.37
340	85.79	177.57	0.36	-0.5925	-0.0377	2754.40	359.44	21.30
341	81.32	194.69	0.14	-0.5165	-0.0563	3529.78	358.93	23.65
342	92.09	201.12	0.17	-0.6598	0.0111	3662.54	349.48	24.76
343	113.24	199.88	0.32	-0.6603	-0.0291	3450.34	148.92	27.93
344	123.35	201.77	0.42	-0.4861	0.1080	3550.23	223.77	31.84
345	149.13	212.31	0.88	-0.4018	0.1025	3961.47	357.88	30.46
346	161.44	219.30	1.26	-0.4446	0.0634	2793.84	354.39	29.24
347	173.03	227.06	0.75	-0.4444	0.0149	2832.17	359.24	25.10
348	180.49	233.78	0.73	-0.4828	-0.0318	3033.71	347.91	19.71
349	81.58	187.33	0.04	-0.5970	-0.0351	2407.40	349.45	18.97
350	82.15	195.03	0.10	-0.4697	-0.0058	2637.04	356.79	20.87
351	83.00	190.17	0.13	-0.6298	-0.0114	2649.57	359.56	19.63
352	84.55	183.19	0.42	-0.5289	0.0547	2378.58	357.02	21.05
353	83.80	181.30	0.24	-0.3884	0.0862	2589.53	359.48	26.85
354	83.29	175.35	0.25	-0.4510	-0.0700	3141.70	356.19	21.86
355	84.75	176.82	0.41	-0.4708	-0.0339	3213.06	356.34	18.46
356	85.30	192.19	0.20	-0.6029	-0.0303	3331.23	341.73	14.71
357	94.81	198.16	0.20	-0.4751	-0.0650	3077.61	14.64	39.47
358	110.32	197.73	0.27	-0.4919	-0.0093	3538.32	346.19	32.00
359	123.65	200.72	0.52	-0.5235	-0.0300	4161.55	356.95	29.75
360	145.39	210.44	0.92	-0.4873	-0.0630	4279.00	358.94	29.41
361	160.89	218.19	1.30	-0.5386	-0.1229	3637.26	357.77	27.28
362	169.07	224.19	0.65	-0.4545	-0.0437	3471.38	357.26	21.18
363	177.62	232.23	0.28	-0.4330	-0.0661	2671.18	359.00	20.33
364	80.24	184.82	0.28	X	X	2605.89	355.30	21.76
365	80.79	184.23	0.34	X	X	2995.96	350.81	26.35
366	82.13	183.63	0.17	X	X	2874.92	359.81	26.30
367	84.18	182.89	0.27	X	X	3198.44	355.01	21.41

368	85.96	184.32	0.34	X	X	2973.23	359.51	21.34
369	89.82	187.58	0.26	X	X	3214.57	357.47	20.89
370	95.97	190.28	0.26	X	X	3469.10	101.47	23.09
371	108.75	193.63	0.27	X	X	3678.18	307.96	23.53
372	118.12	196.13	0.23	X	X	3919.14	277.65	25.58
373	148.06	208.68	0.69	X	X	4256.27	344.12	23.67
374	164.93	218.35	1.01	X	X	4287.31	351.15	30.94
375	168.14	221.57	0.74	X	X	3818.31	358.99	31.32
376	173.54	224.98	0.05	X	X	3441.65	356.79	28.30
377	79.34	183.14	0.04	X	X	3056.43	354.38	25.78
378	79.87	184.68	0.12	X	X	2961.10	344.27	16.17
379	80.64	183.76	0.09	X	X	3601.69	245.19	12.82
380	82.48	183.14	0.28	X	X	3819.84	297.97	20.68
381	84.54	184.27	0.11	X	X	4004.98	311.02	12.71
382	88.47	185.25	0.27	X	X	4179.67	356.18	22.32
383	94.73	187.22	0.33	X	X	3960.90	356.38	29.61
384	105.35	190.51	0.14	X	X	4206.90	357.74	30.10
385	116.81	193.80	0.22	X	X	3787.65	346.97	27.72
386	140.65	204.62	0.60	X	X	3597.38	191.77	11.00
387	159.80	215.13	0.90	X	X	3786.21	301.41	15.16
388	163.00	217.00	0.36	X	X	3907.41	351.03	22.99
389	79.26	179.45	0.08	X	X	4395.50	251.03	29.44
390	79.06	179.06	0.28	X	X	4082.25	354.75	32.54
391	81.48	180.01	0.15	X	X	4190.50	0.00	0.00
392	85.82	181.24	0.23	X	X	4366.83	0.00	0.00
393	90.89	182.48	0.22	X	X	4455.68	0.00	0.00
394	98.74	185.88	0.07	X	X	0.00	0.00	0.00
395	77.54	177.04	0.08	X	X	0.00	0.00	0.00
396	79.28	178.42	0.07	X	X	0.00	0.00	0.00
397	84.34	178.93	0.17	X	X	0.00	0.00	0.00
398	86.85	179.76	0.19	X	X	0.00	0.00	0.00
399	95.52	182.27	0.01	X	X	0.00	0.00	0.00

4. CONCLUSION

The 2014 Jammu and Kashmir floods serve as a stark reminder of the destructive potential of extreme weather events, particularly in ecologically sensitive and topographically complex regions. The spatial and temporal analysis conducted through rainfall interpolation, topographical mapping, and NDWI assessments underscores the multifactorial nature of the disaster.

A dramatic surge in precipitation during September, especially across the south-central districts such as Reasi, Ramban, and Jammu, overwhelmed the natural drainage systems and led to catastrophic inundation. NDWI analysis revealed significant increases in surface water in most districts post-flood, especially in low-lying areas with flatter slopes, reinforcing the impact of topography on water retention.

Conversely, regions like Mirpur showed a slight decline, possibly indicating effective post-event drainage. The combination of intense rainfall, steep slope-induced runoff, urban sprawl, and infrastructural inadequacies contributed to the scale of devastation observed. This study highlights the importance of integrating climatic data with geospatial tools to not only understand the dynamics of such disasters but also to strengthen early warning systems and flood management strategies in vulnerable regions like Jammu and Kashmir.

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